

Daily Discount Decisions for Perishable Food

Forecast tomorrow's sales. Compare supported discounts. Explain when to abstain.

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01 The business question

Can historical sales help a category manager decide **whether to discount a perishable SKU tomorrow, and by how much**, while limiting loss of predicted sales value?

Public observations, no synthetic labels.
FreshRetailNet-50K (Dingdong): 312 retrospectively selected store-SKU series; 2,184 final-period daily cases. Target: observed normalized sales, which can be limited by stockouts.

TIME ORDER (2024) Train from 4 April | Validate 5 expanding weeks, 22 May–25 June | Benchmark 26 June–2 July
One day ahead, repeatedly: only earlier observed days update history. The final benchmark was already viewed during prior development.

02 From history to a supported decision

1	→ 2	→ 3	→ 4	→ 5
Build the context	Learn sales patterns	Select by validation	Score price options	Recommend or abstain
Recent sales, past stockouts, calendar and planned discount. Shift history before computing rolling features.	Fit models on earlier dates. Compare all methods with the same evaluation rows and preprocessing rules.	Choose the smallest mean weekly MAE. Use the final period for a labelled benchmark, not further selection.	Require 3 nearby training days. Predict sales at supported rates, keeping context fixed. Value proxy = rate × predicted sales.	Require 5% sales gain, 95% of reference value and model agreement. Prefer milder near-best discounts; otherwise retain or review.

03 Forecast evidence

METHOD	FINAL-PERIOD MAE	VAL. MAE	SELECTED USING VALIDATION
7-day mean		0.3142	0.3284
Same weekday		0.3998	0.4150
Original Random Forest		0.2923	0.3024
Original CatBoost		0.2896	0.3033
CatBoost MAE		0.2884	0.3027
CatBoost + history/IDs (RMSE)		0.2842	0.2972
CatBoost + history/IDs (MAE)		0.2846	0.2945
Original blend (50/50)		0.2892	0.3013
Proposed blend (25/75)		0.2834	0.2940

MAE is mean absolute prediction error; lower is better. 9 of 17 methods shown; all are in the notebook. Teal marks the validation selection.

Controlled technical test: with history/IDs fixed, aligning CatBoost's loss with MAE is expected to reduce absolute errors. Validation MAE: **0.2972 (RMSE loss) vs. 0.2945 (MAE loss)**. The measured change is predictive, not a causal pricing gain.

04 A traceable discount example

Store 18 · SKU 11 · 2024-06-26

Known history: yesterday's sales **1.400**; 7-day mean **1.257**; yesterday's stockout hours **0**.

PRICE RATE	PRED. SALES	RATE × SALES	HISTORY DAYS
1.00	1.074	1.074	33
0.95	1.240	1.178	10
0.90*	1.291	1.162	12

Decision: 0.90 selected (*). Supported candidate passes model agreement and value floor; mildest near-best sales option.

A rate of 0.90 means 10% off. Alternative-rate predictions are what-if estimates; their outcomes were not observed.

05 Business use and failures

Screen all 2,184 cases: 1,179 recommend a discount scenario; 967 require supply review; 38 other cases need review/abstention. **96.9%** of the 1,217 cases without a recent-stockout flag still suggest discounting. This high rate needs further investigation.

Transfer stress check: 700 cases across 93 new stores / 16 other cities. Proposed vs. original blend: MAE **0.2890 vs. 0.2884**; RMSE **0.4886 vs. 0.4710**. The improvement does not transfer here. We kept the model frozen; no retuning.

Limits: discounts are observational; stockouts do not reveal inventory age or waste. Rate × sales is a proxy, not measured revenue or profit. Pricing gains require intervention evidence.